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27 JANUARY 1964

SUPERSEDING
MIL-M-46363(MU)
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MILITARY SPECIFICATION

MOUNT, PERISCOPE: M113

1. SCOPE

1.1 This specification covers one type of mount known as Mount, Periscope, M113 used to support Periscope XM30 or M30C in the Tank, Combat, Full Tracked, Flame Thrower M67AI or M67A2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 — Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel; Packaging of.

MIL-45208 —Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8634465 —Mount, Periscope, M113.

IEL-8634465 —Index of Inspection Equipment Lists for Mount, Periscope: M113.

LSQAP-8634465 —List of Supplementary Quality Assurance Provisions for Mount, Periscope: M113.

PACKAGING DATA SHEET

MIL-P-14232/ P8634465 -Packaging of Mount, Periscope, M113.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordante with the drawings, material specifications and general specifications forming a part of this specification.

3,2 Fabrication. The mount shall be manufactured in accordance with Drawing F8634-465 Mount, Periscope, M113 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quaIity and performance capabilities.

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3.3 General specifications. The mount shall meet the following requirements of Specification MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Operating temperaturcs. The oldham coupling, double eccentric adjuster as indicated on Drawing 138614774, and all moveable linkage assemblies shall be mechanically operable, and the mount shall meet all the performance requirements herein when exposed and thermally stabilized at ambient temperatures of +150° and −40°.

3.5 Performance.

3.5.1 Level vial. The level vial bubble shall be centered within one graduation line width when the mount is positioned in compliance with the Notes on Sheet 3 of Drawing F863-4467.

3.5.2 Functional operation. With the mount in its normal operating position (adapter plate horizontal and linkage in accordance with Drawing F8634465) the mount shall firmly secure a medium representative of a Periscope, M30 or equal when the locking. handIe of the mount is actuated to the locked position and held in place by the projecting spring loaded button in the collar assembly (Dw'g. D8270628). When the Iocking handle is moved to the unlocked position, the catch assembly (Dwg. B8205659) shall prevent the medium from falling free of the mount.

3.5.2.1 Backlash. Backlash between the Pivot bolt, 8213050 at the end of the linkage and the P1ate, 8247024 on Coupling, 8614811 shall not exceed 0.2 mil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility y for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-1-45208 and the component inspection requirements of Specification MIL-F-13962. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105,

4.2.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926, and List of Supplementary Quality Assurance Provisions, LSQAP-8634465. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

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TABLE I. Classijfcation of defects
Use Inspection Level II of table I with Sampling Plan table II–A of Standard MIL–STD–105.

	Requirement	Test procedure
Critical: None		
Major: 4.0%		
101. Level Vial	3.5.1	4.5.2.1
102. Functional operation.	3.5.2	4.5.2.2
103. Backlash	3.5.2.1	4.5.2.3
Minor: None		

4.2.3 Disposition of nonconforming product. Rejected lots shall be screened for defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests. Three mounts shall be selected at random from each 50 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:.

Control test	Requirement	Teat proccdure
Materials	3.1	Appropriate Dwg & Specifications.
Fabrication	3.2	Visual–As per dwgs,
General Specs	3.3	VisuaL-MIL-F-13926
Operating Temperatures	3.4.1	4.5.1

4.3.1 Control test failure. Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise providecl for by the contract, the contractor shall furnish and maintain all requirecl measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL-I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design, All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the List of Inspection Equipment Numbers, IEL-8634465. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such -as to facilitate routine calibration of the equipment.

4.4.3 Equipment to test level vial, functional operation and backlash. The test equipment shall be so constructed as to permit the periscope mount to be positioned as shown on Drawing F8634465. The test equipment shall also provide a pivot arm that accepts Bolt, 8213050 to form a parallelogram with Arm, 8614785 and Link Assembly, 8213052. The pivot point for this arm shall be established by means of the 3.197 and 31.500 inch

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dimensions shown on Drawing F8634465. The equipment shall also conform to the following:

(a) Level vial. The test equipment shall provide a means of positioning the Arm, 8614785 within a tolerance of plus or minus 0.0005 inch.

(b) Functional operation. The test equipment used to simulate the function of the periscope shall conform to Drawing F8229243.

(c) Backlash. The telescope used in this test shall be equipped with a crosshair reticle, the reticle line width shall not exceed 0.001 inch. The telescope shall be equipped with an adapter to permit mounting on Plate, 8247024. The adapter shall contain provisions for maintaining the telescope optical axis perpendicular to surfaces "x" and "y" of the coupling within plus or minus 60 minutes. The target utilized for this test shall contain acceptance or rejection limitation markings that shall be positioned within an accuracy of 0.1 mil. The target vertical line shall be plumb.

4.5 Test methods and procedure.

4,5.1 Operating temperatures. The condition for temperature cycling for the operability test of the mount at the specified temperature extremes specified in 3.5.1 shall be in accordance with "Test Facilities" of Specification MIL-F-13926. When the mount is thermally stabilized at specified temperatures, it shall be subjected to the tests specified in 4.5.2.1 through 4.5.2.3 inclusive and shall meet the performance requirements at each temperature extreme. The components and assemblies indicated in 3.5.1 shall be subjected to a tactile examination of the low and high temperatures to determine compliance with 3.5.1, Return all components and assemblies to their original position following this test.

4.5.2 Performance. The following tests shall be performed with the aid of special test equipment that conforms to the requirements of 4.4.2.2.

4.5.2.1 Level vial. Position the Arm 8614786 to the basic dimension specified on Sheet 3 of Drawing 8634467 and apply the required static torque to surfaces "x" and "y" of the coupling. The level vial shall meet the requirements of 3.5.1. The verticality of the plane formed by the coupling surfaces "x" and "y" shall be measured with standard measuring equipment and shall meet the requirements specified on Sheet 3 of Drawing F8634467.

4.5.2.2 Functional operation. The special test equipment specified in 4.4,3 shall be used to perform this test. The mount shall meet the retention requirements of 3.5.2 when the device is inserted into the mount and the locking handle is actuated as specified.

4.5.2.3 Backlash. The telescope and target specified in 4.4.3 shall be used to perform this test. Secure the telescope to Plate, 8267024 with its optical axis perpendicular to the plane established by surfaces "x" and "y". Position the target the calibrated distance from the mount with its vertical line plumb and its horizontal line laying in the horizontal plane containing the axis of Coupling, 8614811. Energy to actuate the parallelogram linkage shall applied at the Bolt, 8213050. With backlash removed for direction of travel, establish coincidence of the telescope horizontal reticle line and the horizontal target line image shall not exceed the tolerance specified in 3.5.2.1 as measured on the target. Repeat this test by depressing the telescope line of sight 15 degrees then returning it to the horizontal. Any backlash resulting from this movement shall not exceed the tolerance specified in 3.5.2.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/-P8634465. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use. The periscope mount is intended to maintain the line of sight of Periscope XM30 or M30C in a fixed relationship to the bore axis of the coaxial machine gun and the main armament of the Tank, Combat, Full Tracked, Flame Thrower M67A1 or M67A2 by synchronously transmitting to the periscope line of sight, any angular displacement of the armament.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable Packaging Data Sheet number (see 5.1).
(d) Applicable stock number.

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